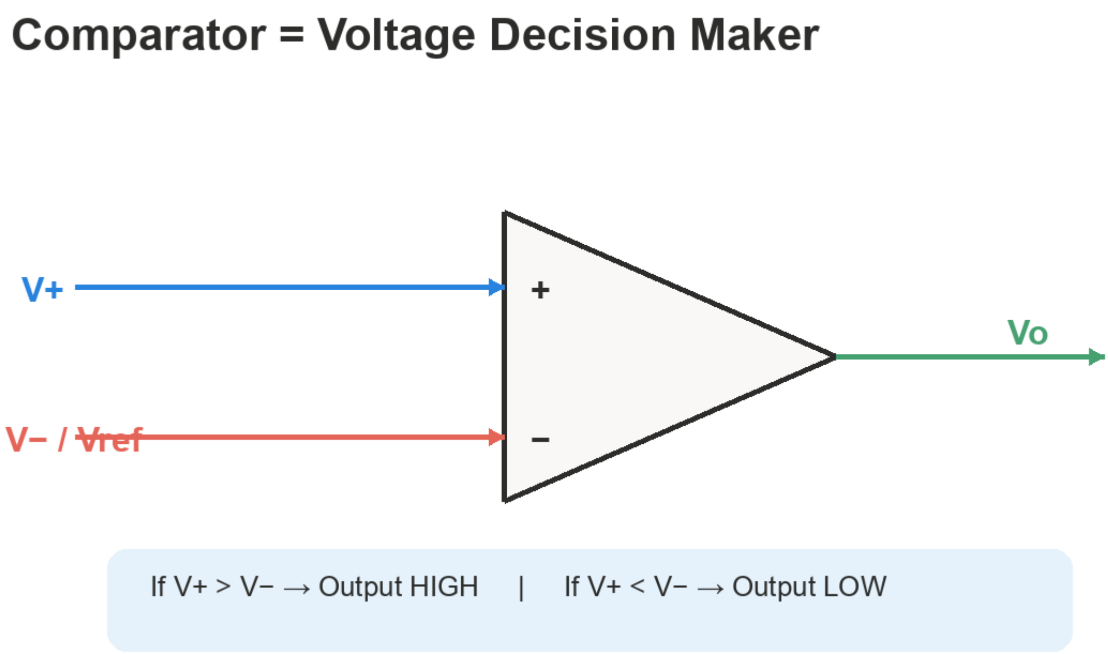
Experiment 04

Comparator and Schmitt Trigger — Diagram Explanation

This PDF explains the practical using diagrams and short simple notes.

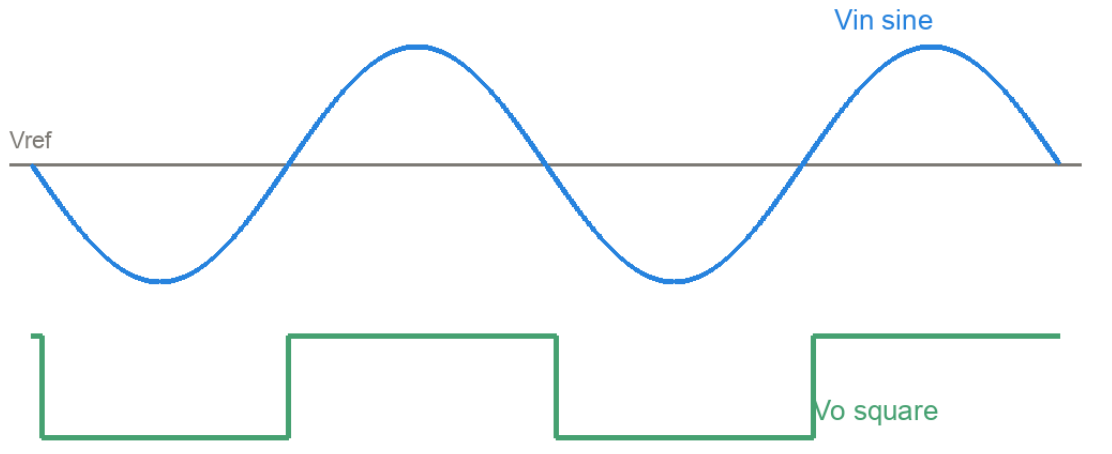
Main idea: comparator = one switching level; Schmitt trigger = two switching levels with hysteresis.



Equation: Vo = A(V+ – V–). Because A is very large, the output jumps HIGH or LOW.

1) Comparator Waveforms

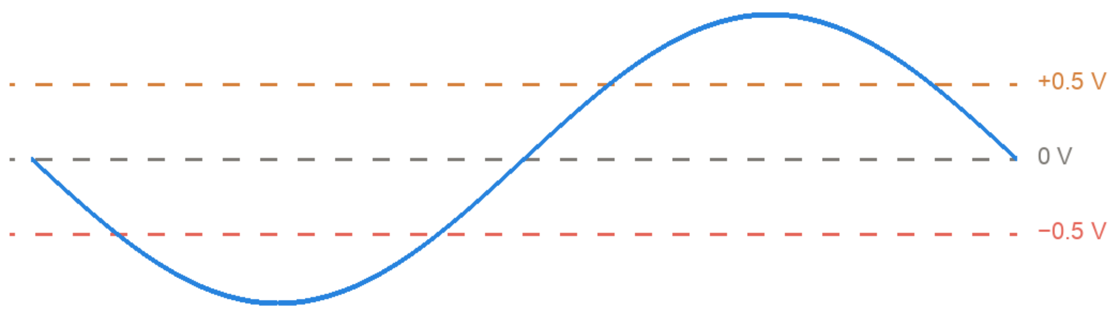
Comparator Waveform: Sine Input → Square Output



Output changes when input crosses the reference level.

When the sine wave crosses the reference level, the comparator output changes state.

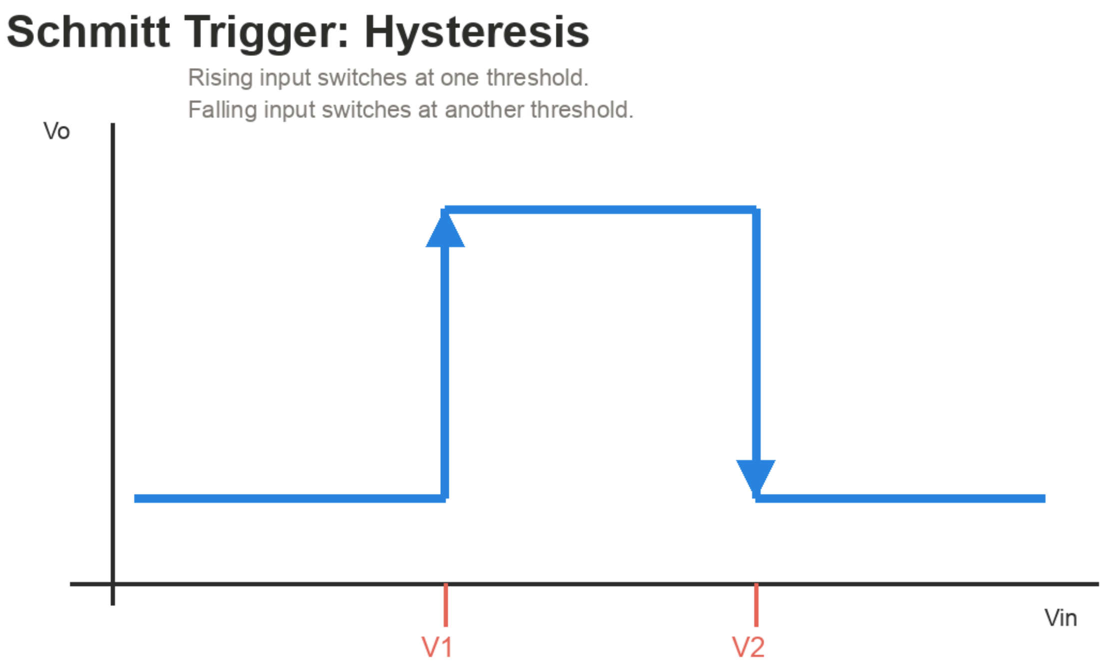
Changing Vref Changes the Switching Point



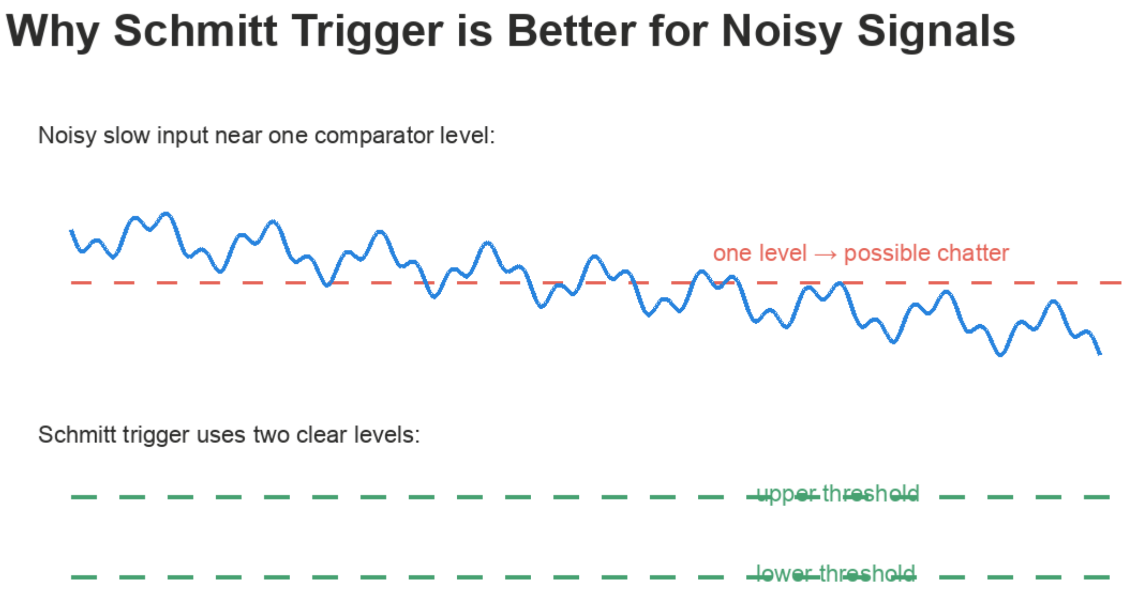
Higher Vref = input must rise more before switching.
Lower Vref = switching happens lower on the sine wave.

Changing Vref moves the switching point. This changes the output timing and pulse width.

2) Schmitt Trigger and Hysteresis



The Schmitt trigger switches at one threshold when input rises and another threshold when input falls.

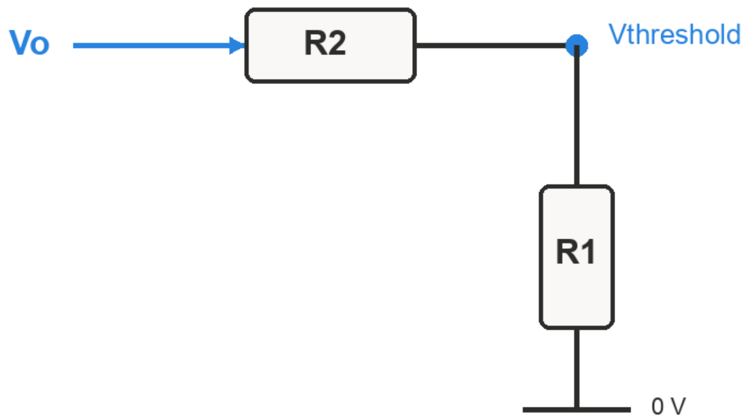


This is why a Schmitt trigger is better for noisy signals.

3) Thresholds and Components

Schmitt Trigger Thresholds from R1 and R2

R1 and R2 decide the threshold and hysteresis width.



$$VT = \left[R1 / (R1 + R2) \right] \times Vo$$

R1 and R2 form a feedback divider. They decide the upper and lower threshold values.

Simplified threshold equation: $VT = [R1 / (R1 + R2)] \times Vo$

Circuit	Important parts	Oscilloscope result
Comparator	Vref, op-amp, input diodes	Sine input becomes square output
Schmitt Trigger	positive feedback, R1, R2, Zener	Clean square output with hysteresis

4) Quick Discussion Answers

Rectifier diodes in Figure 2

They clamp/protect the comparator input so the input difference does not become too large.

Zener diode in Figure 3

It clamps the output near 5.1 V, helping create stable threshold values.

Vref in comparator

Vref is the switching level. Output changes when Vin crosses Vref.

R1 and R2 in Schmitt trigger

They set the feedback ratio, so they decide upper and lower thresholds.

Comparator vs Schmitt trigger

Comparator has one switching level. Schmitt trigger has two levels and better noise immunity.

Op-amp comparator limitation

A 741 op-amp is slower than a real comparator and has saturation recovery delay.

Final memory: Comparator compares. Schmitt trigger compares with memory.